

Disproof of Conjecture 00000001753: spin character values of $2 \cdot S_n$ on even classes

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Abstract

Conjecture 00000001753 asserts that every spin character χ of the double cover $2 \cdot S_n$ satisfies $|\chi(g)| \leq 2^{\lfloor n/4 \rfloor - 1}$ for every element g in an “alternating class”, i.e. lying over an even permutation. This is false. For $n = 5$ the bound is $2^{1-1} = 1$, but the basic spin character χ of $2 \cdot S_5$ (degree 4, irreducible) takes the value ∓ 2 at the element $t_1 t_2$, which lies over the 3-cycle (123) . This holds for both double covers of S_5 . By Schur’s formula the lifted 3-cycle violates the bound for every $n \geq 5$, and the identity violates it for $n = 4$. The $n = 5$ counterexample is formalized in Lean 4 (core library only), using Schur’s presentation of the double covers, an explicit 4-dimensional representation over $\mathbb{Z}[\zeta_8]$, and an explicit irreducibility certificate.

1 The conjecture and how we read it

Definition: The spin characters of $2 \cdot S_n$ are the projective characters of the double cover.

Conjecture: The spin character values on alternating classes satisfy $|\langle \chi, g \rangle| \leq 2^{\lfloor n/4 \rfloor - 1}$; controlled by the log-concavity of spin character degrees, improving the order- n bound $2^{(n/2)-1}$ by the decay factor $2^{\lfloor n/4 \rfloor}$.

The Chinese version states the same inequality. We refute the following clause, which is the only mathematical assertion in the statement (the rest is commentary on why it should hold):

$$|\chi(g)| \leq 2^{\lfloor n/4 \rfloor - 1} \quad \text{for every spin character } \chi \text{ of } 2 \cdot S_n \quad (1)$$

and every $g \in 2 \cdot S_n$ lying over an even permutation.

We read the notation as follows.

- $\langle \chi, g \rangle$ is the value $\chi(g)$ of χ at g (the evaluation pairing). The comparison with “the order- n bound $2^{(n/2)-1}$ ”, which is of the size of the degree $2^{\lfloor (n-1)/2 \rfloor}$ of the basic spin character, shows that character values are meant. Another reading is discussed in Section 5.
- A *spin character* of a double cover $1 \rightarrow \{1, z\} \rightarrow \tilde{G} \rightarrow S_n \rightarrow 1$ is the character of a representation ρ with $\rho(z) = -1$. Equivalently, it is the character of a projective representation of S_n that does not lift to a linear one. We use an *irreducible* spin character, so the counterexample works whether or not reducible ones are allowed.
- An *alternating class* is a conjugacy class of $2 \cdot S_n$ lying over A_n , i.e. a class contained in $2 \cdot A_n$. Our element $t_1 t_2$ lies over a 3-cycle, which is even and not the identity.
- There are two inequivalent double covers $\hat{S}_n$ and $\tilde{S}_n$ for $n \geq 4$ (Section 2). We refute the claim for both.

2 The double covers of S_n

Schur [1] showed that, for $n \geq 4$, S_n has exactly two inequivalent non-split central extensions by a group of order 2. They are given by the presentations with generators $t_1, \dots, t_{n-1}$ and a central involution z :

$$\begin{aligned}\widehat{S}_n : \quad & t_i^2 = 1, \quad (t_i t_{i+1})^3 = 1, \quad (t_i t_j)^2 = z \quad (|i - j| \geq 2); \\ \widetilde{S}_n : \quad & t_i^2 = z, \quad (t_i t_{i+1})^3 = z, \quad (t_i t_j)^2 = z \quad (|i - j| \geq 2).\end{aligned}$$

Each presented group has order $2 \cdot n!$, and $t_i \mapsto (i \ i+1)$, $z \mapsto 1$ defines a surjection onto S_n with kernel $\{1, z\}$ (Schur [1]; see also Hoffman–Humphreys [2, Ch. 2] and Stembridge [3]). Both are denoted $2 \cdot S_n$ in the literature, with varying sign conventions ($2 \cdot S_n^\pm$), so we name them by their relations.

By von Dyck's theorem, a representation ρ of $\widehat{S}_n$ (resp. $\widetilde{S}_n$) with $\rho(z) = -I$ is the same thing as a list of matrices $T_1, \dots, T_{n-1}$ satisfying the relations with z replaced by $-I$. The scalar $-I$ is automatically central and squares to I . An element of the group is $\pm$ a word in the t_i , and the word $t_{i_1} \cdots t_{i_k}$ lies over the permutation $s_{i_1} \cdots s_{i_k}$, where $s_i = (i \ i+1)$.

Remark 1. If $T_1, \dots, T_{n-1}$ satisfy the $\widehat{S}_n$ relations, then $iT_1, \dots, iT_{n-1}$ satisfy the $\widetilde{S}_n$ relations, since $(iT_i)^2 = -T_i^2$, $(iT_i iT_{i+1})^3 = -(T_i T_{i+1})^3$ and $(iT_i iT_j)^2 = (T_i T_j)^2$. A word of even length k gets the factor $i^k = \pm 1$, so on elements lying over even permutations the two characters agree up to sign, and so do their absolute values.

3 The counterexample at $n = 5$

Let $\xi_1, \dots, \xi_5 \in M_4(\mathbb{C})$ be pairwise anticommuting involutions: $\xi_a \xi_b + \xi_b \xi_a = 2\delta_{ab}I$. For example, with the Pauli matrices X, Y, Z , take

$$\xi_1 = X \otimes I, \quad \xi_2 = Y \otimes I, \quad \xi_3 = Z \otimes X, \quad \xi_4 = Z \otimes Y, \quad \xi_5 = Z \otimes Z.$$

For $v \in \mathbb{R}^5$ put $\xi(v) = \sum_a v_a \xi_a$; then $\xi(v)\xi(w) + \xi(w)\xi(v) = 2(v \cdot w)I$. Let $\alpha_i = (e_i - e_{i+1})/\sqrt{2}$ for $i = 1, \dots, 4$, so that $\alpha_i \cdot \alpha_i = 1$, $\alpha_i \cdot \alpha_{i+1} = -\frac{1}{2}$ and $\alpha_i \cdot \alpha_j = 0$ for $|i - j| \geq 2$. Define

$$T_i = \xi(\alpha_i) = \frac{1}{\sqrt{2}}(\xi_i - \xi_{i+1}) \quad (i = 1, \dots, 4).$$

Lemma 2. *The T_i satisfy the relations of $\widehat{S}_5$ with $z \mapsto -I$.*

Proof. $T_i^2 = (\alpha_i \cdot \alpha_i)I = I$. For $|i - j| \geq 2$ the anticommutator is $2(\alpha_i \cdot \alpha_j)I = 0$, so $T_i T_j = -T_j T_i$ and $(T_i T_j)^2 = -T_i^2 T_j^2 = -I$. For $j = i + 1$ write $u = \alpha_i$, $v = \alpha_{i+1}$ and $P = \xi(u)\xi(v)$. From $\xi(u)\xi(v) + \xi(v)\xi(u) = -I$ we get $P + P^{-1} = P + \xi(v)\xi(u) = -I$, so $P^2 + P + I = 0$ and hence $P^3 = I$. $\square$

So $t_i \mapsto T_i$ defines a spin representation ρ of $\widehat{S}_5$, and $t_i \mapsto iT_i$ one of $\widetilde{S}_5$. Let $\chi = \text{tr } \rho$; note $\chi(z) = \text{tr}(-I) = -4 = -\chi(1)$.

Lemma 3. *ρ is irreducible. In fact the matrices $\rho(w)$, w a word in the t_i , span $M_4(\mathbb{C})$.*

Proof. The α_i span the 4-dimensional space $V = \{x \in \mathbb{R}^5 : \sum x_a = 0\}$, so the algebra A generated by the T_i is the image of the complex Clifford algebra $\text{Cl}(V)_\mathbb{C}$ under ξ . Since $\dim V = 4$ is even, $\text{Cl}(V)_\mathbb{C} \cong M_4(\mathbb{C})$ is a simple algebra of dimension 16. The map to $M_4(\mathbb{C})$ is nonzero, hence injective, hence surjective by dimension. An invariant subspace of ρ is invariant under $A = M_4(\mathbb{C})$, so it is 0 or $\mathbb{C}^4$. Concretely, the 16 products $T_{i_1} \cdots T_{i_k}$ with $i_1 < \cdots < i_k$ form a basis of $M_4(\mathbb{C})$ (they are the images of a basis of $\text{Cl}(V)$). $\square$

Hence χ is an irreducible spin character of $\widehat{S}_5$ of degree 4, the *basic spin character*. (Its values, computed in `verify.py`, are ± 4 on ± 1 , ± 2 on lifts of 3-cycles, ± 1 on lifts of 5-cycles, and 0 elsewhere, so $\langle \chi, \chi \rangle = (2 \cdot 16 + 40 \cdot 4 + 48 \cdot 1)/240 = 1$.)

Theorem 4. *Let $g = t_1 t_2$. Then g lies over the even permutation $s_1 s_2 = (1\ 2\ 3)$, and $\chi(g) = -2$ for $\widehat{S}_5$ (and $+2$ for $\widetilde{S}_5$). So $|\chi(g)| = 2 > 1 = 2^{\lfloor 5/4 \rfloor - 1}$, and (1) fails for $n = 5$ and both double covers.*

Proof. $s_1 s_2 = (1\ 2)(2\ 3) = (1\ 2\ 3)$ is a 3-cycle, an even permutation. Next,

$$T_1 T_2 = \frac{1}{2}(\xi_1 - \xi_2)(\xi_2 - \xi_3) = \frac{1}{2}(\xi_1 \xi_2 - \xi_1 \xi_3 - I + \xi_2 \xi_3).$$

For $a \neq b$, $\text{tr}(\xi_a \xi_b) = \text{tr}(\xi_b \xi_a) = -\text{tr}(\xi_a \xi_b)$, so $\text{tr}(\xi_a \xi_b) = 0$. Hence $\chi(t_1 t_2) = -\frac{1}{2} \text{tr } I = -2$. For $\widetilde{S}_5$ the element $t_1 t_2$ acts by $(iT_1)(iT_2) = -T_1 T_2$, with trace $+2$. $\square$

The identity element already gives $|\chi(1)| = 4 > 1$. We use the lifted 3-cycle so that the counterexample does not depend on whether the trivial class counts as an “alternating class”.

4 All n

The construction above works for every n : take n anticommuting involutions ξ_a of size $2^{\lfloor n/2 \rfloor}$ and $T_i = (\xi_i - \xi_{i+1})/\sqrt{2}$. Schur’s formula for the basic spin character $\langle n \rangle$ (Schur [1]; see also [3] and [2]) reads as follows. Its degree is $2^{\lfloor (n-1)/2 \rfloor}$. On the lifts of a permutation of cycle type π with all parts odd it takes the values $\pm 2^{\lfloor (\ell(\pi)-1)/2 \rfloor}$, where $\ell(\pi)$ is the number of cycles (fixed points included). It vanishes on all other classes, except that for n even the two associate characters are nonzero on the lifts of n -cycles, with absolute value $\sqrt{n/2}$.

For a 3-cycle, $\pi = (3, 1^{n-3})$ and $\ell(\pi) = n - 2$, so the value is $\pm 2^{\lfloor (n-3)/2 \rfloor}$. Now

$$\left\lfloor \frac{n-3}{2} \right\rfloor \geq \frac{n-4}{2} > \frac{n}{4} - 1 \geq \left\lfloor \frac{n}{4} \right\rfloor - 1 \quad (n \geq 5),$$

where the strict inequality is equivalent to $n > 4$. So for *every* $n \geq 5$ the lifted 3-cycle violates (1), in both double covers. For $n = 4$ the basic spin characters have degree $2 > 1 = 2^0$, so the identity violates it (the lifted 3-cycle has value ± 1 and does not). Thus (1) fails for every $n \geq 4$, and in particular for all large n .

`verify.py` confirms Schur’s formula on all even classes for $n = 4, 5, 6, 7$ by computing the groups ($|2 \cdot S_n| = 48, 240, 1440, 10080$) and their characters.

5 Other readings

1. **Which double cover.** Both $\widehat{S}_5$ and $\widetilde{S}_5$ are refuted (Theorem 4), in Lean as well.
2. **Classes of $2 \cdot A_n$.** The element $t_1 t_2$ lies in $2 \cdot A_5$, so it is in an alternating class under any reading (whether a class of $2 \cdot S_5$ inside $2 \cdot A_5$, or a class of the group $2 \cdot A_5$ itself). The character value does not depend on this choice.
3. **Characters of $2 \cdot A_n$.** If “spin character” meant a faithful irreducible character of $2 \cdot A_n$, the claim still fails at $n = 5$. Restricted to $2 \cdot A_5 \cong \text{SL}(2, 5)$, the basic spin character splits as $\psi_+ + \psi_-$ with $\deg \psi_{\pm} = 2$. On a lift of a 5-cycle, $\psi_{\pm} = (1 \pm \sqrt{5})/2$, and $|(1 + \sqrt{5})/2| > 1$; also $\psi_{\pm}(1) = 2 > 1$. (`verify.py`, Part D. It uses the matrix $s = \xi_1 + \dots + \xi_5$, which commutes with $\rho(2 \cdot A_5)$ and has $s^2 = 5$, so that $\psi_{\pm}(g) = \frac{1}{2}(\chi(g) \pm \text{tr}(gs)/\sqrt{5})$.)

4. **“Alternating classes” read as odd classes.** The claim still fails. For $n = 4$ the two associate basic spin characters have $|\chi_{\pm}| = \sqrt{2} > 1$ on the lifts of a 4-cycle (`verify.py`, Part D, exact). For $n = 5$, Schur’s formula gives the associate pair of degree 6 (labelled by $(4, 1)$) the absolute value $\sqrt{4 \cdot 1/2} = \sqrt{2} > 1$ on the lifts of cycle type $(4, 1)$ (cited, not computed).
5. **“For large n ”.** The claim fails for every $n \geq 5$ (Section 4).
6. **Strict or non-strict.** $|\chi(g)| = 2 > 1$ violates both $\leq$ and $<$.
7. **Reducible spin characters.** Our χ is irreducible, so it is a spin character under either convention.
8. **$\langle \chi, g \rangle$ as an inner product.** If $\langle \chi, g \rangle$ meant the inner product $\langle \chi, \mathbf{1}_C \rangle = |C| \chi(g) / |G|$ with the indicator of the class C of g , the statement would be trivially true for $n \geq 4$. Indeed $|C| |\chi(g)| / |G| = |\chi(g)| / |C_G(g)| \leq |C_G(g)|^{-1/2} \leq 1$ by column orthogonality ($|\chi(g)|^2 \leq |C_G(g)|$). This makes the bound $2^{\lfloor n/4 \rfloor - 1}$ and the comparison with “the order- n bound $2^{(n/2)-1}$ ” meaningless, so we regard it as unintended. We make no claim under this reading.

6 Lean formalization

The folder `lean4/` is a Lean 4.19.0 project using only the core library. It builds in about 40 seconds with no `sorry`, no `native_decide` and no added axioms. All displayed theorems depend on no axioms at all. Finite computations use `decide` and `decide +kernel`.

Objects.

- **Z8:** the ring $\mathbb{Z}[\zeta]$, $\zeta = e^{\pi i/4}$, as quadruples $a + b\zeta + c\zeta^2 + d\zeta^3$ with $\zeta^4 = -1$. It has complex conjugation `conj` ($\zeta \mapsto -\zeta^3$) and `normSq`(x) = $x\bar{x} = |x|^2$. It is a subring of $\mathbb{C}$.
- Matrices are lists of rows, with product `mmul`, `trace` and `ident`. `eval d T w` is the matrix $T_{w_0} T_{w_1} \cdots$ of a word w .
- `wordPerm n w` is the permutation $s_{w_0} s_{w_1} \cdots$ (one-line notation), and `IsEven` is evenness via the number of inversions.
- `Cover = {hat, tilde}`. `IsSpinRep c n d T`: T is a list of $n-1$ square $d \times d$ matrices ($d > 0$) satisfying Schur’s relations for the cover c with $z \mapsto -I$. By von Dyck’s theorem this is exactly a spin representation of the presented group.
- `SpansMatrices d T`: for all $a, b < d$ there are $0 \neq N \in \mathbb{Z}[\zeta]$ and a $\mathbb{Z}[\zeta]$ -combination of matrices of words equal to NE_{ab} . Then the words span $M_d(\mathbb{C})$, so the representation is irreducible over $\mathbb{C}$ (as in Lemma 3).
- `Claim c n`: for all d and T with `IsSpinRep c n d T` and `SpansMatrices d T`, for every word w in the generators with `wordPerm n w` even, and for every $q \in \mathbb{Z}$, if `normSq(tr eval d T w) = q` then $q \leq 4^{\lfloor n/4 \rfloor - 1}$.

`Claim c n` is implied by (1) for the cover c . It only quantifies over irreducible spin representations realized over $\mathbb{Z}[\zeta]$, over elements given by words, and only constrains $|\chi(g)|^2$ when it is a rational integer. So `¬Claim` refutes (1).

The data. T5 is the representation of Section 3, written in a basis of a $\rho(\widehat{S}_5)$ -stable $\mathbb{Z}[\zeta]$ -lattice, so that all entries lie in $\mathbb{Z}[\zeta]$. For $\widetilde{S}_5$ the generators are $i \cdot \mathbf{T5}$. The certificate `irrCert` expresses each $5E_{ab}$ as a $\mathbb{Z}[\zeta]$ -combination of the 16 words $t_{i_1} \cdots t_{i_k}$ ($i_1 < \cdots < i_k$). The denominator 5 is the discriminant of the A_4 root lattice.

Theorems.

- `spinRep c`: Schur's relations hold for both covers.
- `spans c`: the irreducibility certificate checks for both covers.
- `t1t2_perm, t1t2_even`: $t_1 t_2$ lies over $[1, 2, 0, 3, 4]$, the 3-cycle $(1\ 2\ 3)$, which is even.
- `chi_t1t2_hat, chi_t1t2_tilde, normSq_t1t2`: $\chi(t_1 t_2) = -2$, resp. 2 , and $|\chi(t_1 t_2)|^2 = 4$.
- `chi_z`: $\chi(z) = -4 = -\chi(1)$.
- The main theorem `conjecture_00000001753_false` proves $\neg \text{Claim } c\ 5$ for each cover c . The theorem `conjecture_00000001753_false.both` states both at once.
- `perm_relations`: $t_i \mapsto s_i, z \mapsto 1$ respects the relations, so the projection to S_5 is well defined.
- `z_ne_one` and `perm_words_distinct`: $\rho(z) = -I \neq I$, and 120 words lie over 120 distinct permutations. Hence the presented group has at least $2 \cdot 120 = 240$ elements, since its image in S_5 has 120 elements and the kernel contains $1 \neq z$.

What is cited rather than formalized. Lean does not prove Schur's theorem that the presented groups have order exactly $2 \cdot 5!$, i.e. that they *are* the double covers $2 \cdot S_5$. (`verify.py` enumerates the image group of T5: 240 elements, kernel $\{\pm I\}$ over S_5 .) Schur's formula for general n (Section 4) and the odd-class value at $n = 5$ are cited. Abstract group theory (von Dyck, Burnside-type spanning) is used in the interpretation of the definitions, as explained above.

7 Independent check

`verify.py` uses only the Python standard library and runs in a few seconds.

- It parses T5 from `Main.lean` and checks the relations of both covers exactly over $\mathbb{Z}[\zeta]$. It enumerates the generated groups (240 elements each), checks that the projection to S_5 is well defined with kernel $\{\pm I\}$, and computes $\langle \chi, \chi \rangle = 1$ (irreducibility by a different method than Lean's certificate) and $\chi(z) = -4$. It lists the values on all even classes and checks $\chi(t_1 t_2) = \mp 2$.
- It builds the Clifford model of Section 3 exactly over $\mathbb{Q}(\zeta)$ and repeats the checks: $|G| = 240$, $\langle \chi, \chi \rangle = 1$, $\chi(t_1 t_2) = -2$.
- For $n = 4, 5, 6, 7$ it checks the relations exactly. It enumerates the group in $\mathbb{F}_p$ with $p = 10^9 + 9 \equiv 1 \pmod{8}$, via the ring map $\mathbb{Z}[\zeta, \frac{1}{2}] \rightarrow \mathbb{F}_p$, and finds $|G| = 2 \cdot n!$ with kernel of order 2. It also checks $\langle \chi, \chi \rangle = 1$ (n odd) or 2 (n even, the sum of the two associates), and that all even-class values agree with Schur's formula. The lifted 3-cycle has $|\chi| = 2, 2, 4$ for $n = 5, 6, 7$, against the bound 1. For $n = 4$ the identity violates the bound.

- D. It checks the other readings: odd classes at $n = 4$ ($|\chi_{\pm}|^2 = 2$, exact) and the characters of $2 \cdot A_5$ ($\psi_{\pm} = (1 \pm \sqrt{5})/2$).
- E. It checks the exponent inequality of Section 4 for $5 \leq n < 10^4$.

References

- [1] I. Schur, *Über die Darstellung der symmetrischen und der alternierenden Gruppe durch gebrochene lineare Substitutionen*, J. Reine Angew. Math. **139** (1911), 155–250.
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- [3] J. R. Stembridge, *Shifted tableaux and the projective representations of symmetric groups*, Adv. Math. **74** (1989), 87–134.